

Structure and Obstructions in the Rule 30 Eventual-Periodicity Problem

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September 2026

Abstract

Wolfram’s Prize Problem 1 asks whether the center column of the lone-seed Rule 30 orbit always remains non-periodic. We do not resolve it, and we say so at the outset: Problems 1, 2 and 3 remain open. What we contribute is a structured account of the boundary between what is provable and what is not for this problem. First, we settle the entire eventual-period-*one* subcase for every nonzero finite configuration, a class strictly broader than the lone seed: the two constant-trace fibers are computed exactly, neither meets the finitely supported configurations away from the origin, and the sharp constant-prefix horizon is $w + 2$ for support radius w . Second, we delimit the eventual-period-*two* case from two sides: a mechanized ω -automaton periodicity ladder, for which we prove three separate limitation theorems (its verdict is uniform in right depth, in left depth, and in the diagonal bound, so none of its levers can close the case); and an exhaustive forced-continuation census, which we carry to source length $n = 24$ exhaustively and $n = 30$ by a verified fast method, finding that the exclusion is decided not by a thinning population but by a bounded handful of extremal forcing states. Third, and this is the paper’s methodological purpose, we give the first structured register of the attacks that fail on this problem: ninety-four distinct routes, organized into nine obstruction classes, each with its exact kill reason, and all unified by a single fact — Rule 30’s OR nonlinearity is the only property separating it from the additive Rule 90, whose center column *is* eventually periodic, so any argument that does not use that nonlinearity cannot succeed. The register is the map the field currently lacks. An exhaustive per-route compendium and the full computational archive accompany the paper as a supplement and a data deposit.

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1 Introduction

Rule 30 is the binary cellular automaton on $\{0, 1\}^{\mathbb{Z}}$ with local rule

$$F(x)_i = x_{i-1} \oplus (x_i \vee x_{i+1}), \quad (1)$$

left permutive: fixing x_i, x_{i+1} , the value $F(x)_i$ is a bijection of x_{i-1} .

Wolfram poses three problems about the center column of the lone-seed orbit and offers \$10,000 for each [7]. Problem 1 asks “Does the center column always remain non-periodic?”; Problem 2, “Does each color of cell occur on average equally often in the center column?”; and Problem 3, “Does computing the n th cell of the center column require at least $O(n)$ computational effort?” A solution is a proof over the traditional axioms of standard mathematics. The column is verified non-periodic only through the first 10^9 steps; as Wolfram notes, “there could be a trillion-step transient, after which it’s periodic,” so no finite check can settle Problem 1 — a limit we return to as obstruction H (Section 5). We address Problem 1 throughout, Problem 2 through the period-two thread, and Problem 3 only in the register.

Kopra records the broader finite-configuration version of Problem 1 as open [4]. Jen proved a finite orbit has at most one eventually periodic column [2, Thm. 2b][1, Prop. 3]; Kopra proved the width-two column is never eventually periodic but left width one open, and showed his technique cannot reach it [3].

This paper is organized around one fact that we make load-bearing throughout. Rule 90, $F(x)_i = x_{i-1} \oplus x_{i+1}$, is also left permutive, yet its lone-seed center column *is* eventually periodic (identically zero off the initial step). Hence no argument that uses only left permutivity, additivity, or any property Rule 90 shares can decide Problem 1: it would prove a false statement about Rule 90. The single property Rule 30 has that Rule 90 lacks is the OR in (1), which latches. We call the requirement that an argument depend on this latch the *Rule 90 filter*. Every positive result below passes it; every failure catalogued in Part IV either violates it or hits one of eight further walls.

What is proved. Part I settles eventual period one (Section 2): both constant-trace fibers are computed in closed form, the finite configurations meet them only at the origin, and the horizon is sharp. Part III proves three limitation theorems for the period-two ladder (Section 4). These are honest negative results: they say a specific, natural, mechanized method cannot close period two, and why.

What is measured, and what is conjectured. Part II (Section 3) reduces period-two exclusion to a finite-word statement, carries the exhaustive census to $n = 24$ (and a verified fast method to $n = 30$), and reports a structural phenomenon: near extinction the surviving witnesses collapse onto $O(1)$ extremal forcing states sharing a long common suffix and a single failure symbol. We state this as a conjecture with its supporting data, never as a theorem.

What fails, and why. Part IV (Section 5) is the register. Nine obstruction classes account for the ninety-four routes attempted across this project. We give each class, the routes it kills, the kill-reason type, and its Rule 90 status. The exhaustive per-route list is the accompanying supplement.

Scope. Part V (Section 6) states precisely what is and is not reached, against the published state of the art, and lists the concrete open leads. We claim no progress on the prize problems themselves.

2 Part I: Finite configurations cannot generate a constant trace

This part is lifted, statement and proof, from the standalone note `zero-tail-note.tex`, which carries the full hostile-audited derivation, the Lean and SMT certificates, and the reproducibility appendix; only the two figures and the eight results are reproduced here.

We write $c_t := \text{Tr}(x)_t = F^t(x)_0$, $R_j := x_j$, $L_j := x_{-j}$ ($j \geq 0$), and let $\mathcal{F}$ denote the finitely supported configurations. The trace sees only column 0, but under a left-permutive rule a discrepancy left of the origin propagates rightward along a diagonal and must eventually reach it. The classification rests on this one fact, which uses only locality and left permutivity, not the specific form of (1).

Lemma 2.1 (Triangular uniqueness). *If two configurations agree at every coordinate $i \geq 0$ and have the same central trace, then they are equal.*

Proof. If $-n$ is their rightmost differing coordinate, locality preserves agreement to its right through time n . Left permutivity preserves the discrepancy along $(-n, 0), (-n+1, 1), \dots, (0, n)$, contradicting equal traces. In its finite form, if the rows agree on $i \geq 0$ and their traces agree for $0 \leq t \leq r$, the same induction forces agreement at $-r, -r+1, \dots, -1$. $\square$

Uniqueness alone does not say what the forced left half looks like. The next result computes it: the rows with zero trace form one invariant class $\mathcal{C}_m$ for each position m of the first right-hand one, each class is carried to its predecessor by F , and the left half is a prefix-OR of the right. Figure 1(a) is a member of $\mathcal{C}_3$.

Theorem 2.2 (Zero-trace fiber). *For a prescribed right half with $R_0 = 0$, the zero-trace-compatible left half is unique. If the positive half is nonzero and $m = \min\{j \geq 1 : R_j = 1\}$, the forced left half is $L_j = 0$ ($j < m$), $L_m = 1$, $L_j = j \bmod 2$ ($j > m$), with R_j for $j > m$ arbitrary. If instead the positive half is zero, the unique row is zero. Equivalently, in prefix-OR form,*

$$L_{2k+1} = \bigvee_{j=1}^{2k+1} R_j \quad (k \geq 0), \quad (2)$$

$$L_{2k} = R_{2k} \wedge \neg \bigvee_{j=1}^{2k-1} R_j \quad (k \geq 1). \quad (3)$$

Proof. For $m \geq 1$, let $\mathcal{C}_m$ consist of rows satisfying these conditions. One Rule 30 step gives

$$x'_0 = L_1 \oplus R_1, \quad R'_j = R_{j-1} \oplus (R_j \vee R_{j+1}), \quad L'_j = L_{j+1} \oplus (L_j \vee L_{j-1}). \quad (4)$$

Suppose $m > 1$. These identities give $x'_0 = 0$, $R'_j = L'_j = 0$ ($j < m-1$), $R'_{m-1} = L'_{m-1} = 1$, $L'_m = m \bmod 2$. For $j > m$, including $j = m+1$ because $L_m = 1$, the OR in the last update is one, so $L'_j = (1 - (j \bmod 2)) \oplus 1 = j \bmod 2$. Thus $F(\mathcal{C}_m) \subseteq \mathcal{C}_{m-1}$. For $m = 1$, the alternating left half is fixed, $x'_0 = 1 \oplus 1 = 0$, and $R'_1 = 0 \oplus (1 \vee R_2) = 1$ independently of the right tail; hence $F(\mathcal{C}_1) \subseteq \mathcal{C}_1$. Every $\mathcal{C}_m$ therefore realizes the zero trace. A nonzero right half selects a realization by its first one, and Lemma 2.1 makes it unique; a zero right half forces a zero left half by the same lemma against the zero row. $\square$

The other constant trace has a simpler fiber: one row per right half, and the left half no longer depends on which right half it is.

Theorem 2.3 (All-one fiber). *For a prescribed right half with $R_0 = 1$, the left half compatible with an identically-one trace is unique and independent of the right half: $L_j = 1 \iff j$ is positive and even.*

Proof. Write $L_0 := x_0 = R_0 = 1$, so the third identity of (4), $L'_j = L_{j+1} \oplus (L_j \vee L_{j-1})$, applies at every $j \geq 1$. Under the checkerboard pattern it gives $0 \oplus (1 \vee 0) = 1$ for even j and $1 \oplus (0 \vee 1) = 0$ for odd j , so the left half is fixed by F whatever the right half is. At the origin, $F(x)_0 = L_1 \oplus (L_0 \vee R_1) = 0 \oplus 1 = 1$ independently of R_1 , so the trace is identically one and $L_0 = 1$ is restored at every step. Lemma 2.1 makes the left half unique. $\square$

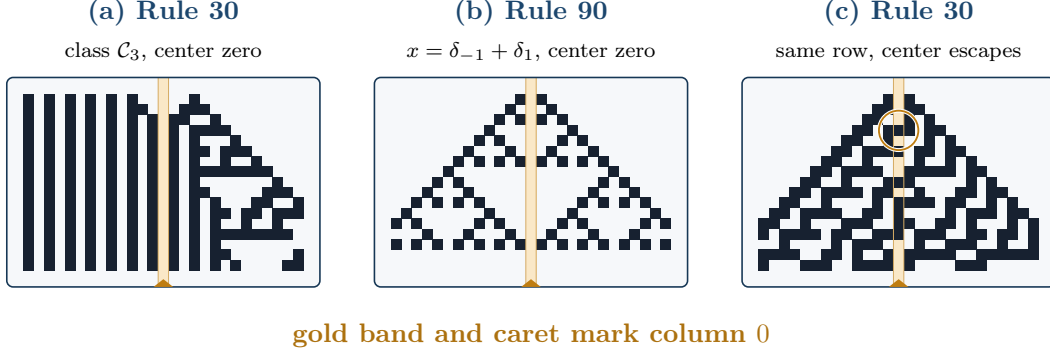


Figure 1: Time runs downward. In (a) the forced left tail alternates, hence is infinite, which is what Corollary 2.4 exploits. Panels (b) and (c) run one finite row under both rules; the difference is the OR latch in (1).

Every nonzero row in Theorem 2.2 carries ones arbitrarily far to the left, so none of them is finitely supported. That is the whole of the next statement, once one checks that a finite orbit cannot become the zero row.

Corollary 2.4. $\text{Tr}^{-1}(0^{\mathbb{N}}) \cap \mathcal{F} = \{0\}$. Moreover, no column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually zero.

Proof. Every nonzero row in Theorem 2.2 has ones at all sufficiently large odd left depths. If a column of a finite orbit were zero from time T onward, translation would make $F^T(x)$ a finite zero-trace row. It is nonzero: if $[a, b]$ is the support interval of a nonzero finite row, its next row has ones at $a - 1$ and $b + 1$ because Rule 30 maps 001 and 100 to one. $\square$

Corollary 2.4 is not a consequence of left permutivity alone: Rule 90 is also left permutive, yet the finite row with ones at ± 1 keeps a zero center forever (Figure 1(b,c)). The Rule-30-specific step is the OR latch $R'_1 = 1 \vee R_2 = 1$ in the proof of Theorem 2.2, which has no analogue in an additive rule.

The same latch upgrades the corollary from one constant to both, with no further machinery. Theorem 2.3 proves the second half again and independently, since the checkerboard pattern has ones at every positive even depth; the argument below is shorter and is the one to quote.

Corollary 2.5. No column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually constant.

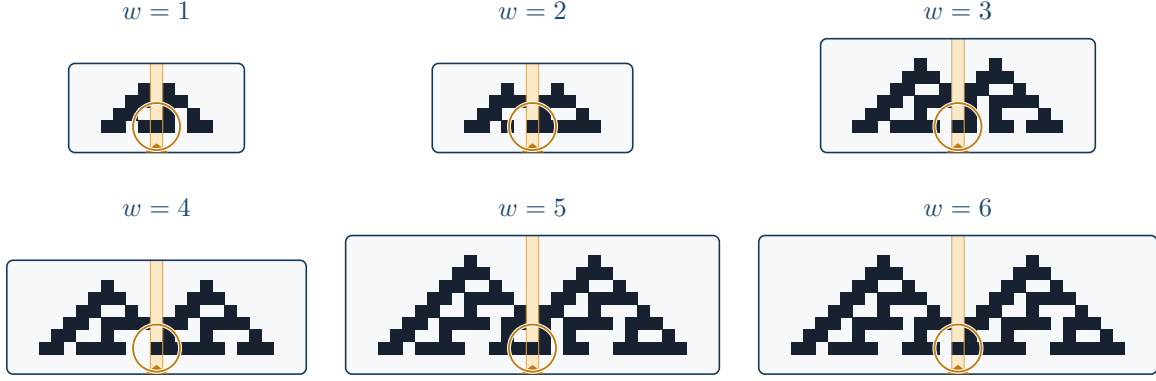
Proof. The eventually-zero case is Corollary 2.4. Suppose instead some column is eventually one; translating, take it to be column 0, so $c_t = 1$ for all $t \geq T$. Applying (1) at the origin gives $l_t = c_{t+1} \oplus (c_t \vee r_t)$, and $c_t = 1$ collapses the disjunction, so $l_t = 1 \oplus c_{t+1}$. For $t \geq T$ we also have $c_{t+1} = 1$, hence $l_t = 0$. Column -1 is then zero from time T onward, which Corollary 2.4 forbids. $\square$

Corollary 2.4 says a finite row's center cannot stay zero forever. How long it can stay zero, as a function of the support radius, is exactly one step past the light cone of the first odd depth beyond that radius. For $w \geq 1$ let $\mathcal{S}_w = \{x \neq 0 : \text{supp}(x) \subseteq [-w, w], x_0 = 0\}$, and for $x \in \mathcal{S}_w$ let $H(x) = \max\{h : F^t(x)_0 = 0 \text{ for } 0 \leq t \leq h\}$.

Theorem 2.6 (Sharp finite horizon). Let $w \geq 1$. Then

$$\max_{x \in \mathcal{S}_w} H(x) = 2 \left\lceil \frac{w}{2} \right\rceil, \quad \#\left\{x \in \mathcal{S}_w : H(x) = 2 \left\lceil \frac{w}{2} \right\rceil\right\} = 2^w - 1. \quad (5)$$

Proof. Let q be the least odd integer greater than w , so $q - 1 = 2\lceil w/2 \rceil$. If the positive right half is nonzero, a trace zero through time q , compared with its zero-trace completion, would force by finite-prefix uniqueness and the prefix-OR identity $x_{-q} = \bigvee_{j=1}^q R_j = 1$, contradicting $q > w$. If that half



circled cell = first nonzero center, at time $2\lceil w/2 \rceil + 1$

Figure 2: One extremizer of Theorem 2.6 per radius, truncated from the zero-trace completion at left depth w . The horizon is flat across each pair $w = 2k - 1, 2k$, which is the parity effect in the statement.

is zero, comparison with the zero row makes every nonzero left cell visible by time w . Thus $H(x) \leq q - 1$. Conversely, choose any of the $2^w - 1$ nonzero words $(R_1, \dots, R_w)$, install the prefix-OR pattern through left depth w , and zero all other cells. This row agrees with its infinite zero-trace completion through the light cone of time $q - 1$; when w is odd, the extra depth $w + 1$ is even and forced to zero. The missing odd cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers. $\square$

The one-trace horizon is the same computation against the checkerboard fiber, and the parities exchange roles: ones sit at even depths, so the binding depth is the least *even* one past the radius, and $R_0 = 1$ already makes the row nonzero, which is why no extremizer is lost here.

Theorem 2.7 (Sharp finite horizon, one). *Let $w \geq 1$, let $\mathcal{T}_w = \{x : \text{supp}(x) \subseteq [-w, w], x_0 = 1\}$, and for $x \in \mathcal{T}_w$ let $H_1(x) = \max\{h : F^t(x)_0 = 1 \text{ for } 0 \leq t \leq h\}$. Let q be the least even integer greater than w . Then $\max_{x \in \mathcal{T}_w} H_1(x) = q - 1$ and $\#\{x \in \mathcal{T}_w : H_1(x) = q - 1\} = 2^w$.*

Proof. A trace one through time q , compared with the all-one completion of Theorem 2.3, would force by finite-prefix uniqueness $x_{-q} = 1$, contradicting $q > w$; thus $H_1(x) \leq q - 1$. Conversely, choose any of the 2^w words $(R_1, \dots, R_w)$, install the checkerboard pattern through left depth w , and zero all other cells. This row agrees with its all-one completion through the light cone of time $q - 1$; when w is even, the extra depth $w + 1$ is odd and forced to zero. The missing even cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers. $\square$

Combining the two horizons, the sharp law over both constants is flat in the parity that splits each one separately.

Corollary 2.8. *Let $w \geq 1$ and let $x \neq 0$ have $\text{supp}(x) \subseteq [-w, w]$. The longest constant prefix of the central trace of x has length at most $w + 2$, and exactly 2^w configurations attain length $w + 2$ when w is even, while exactly $2^w - 1$ attain it when w is odd.*

Proof. For $x_0 = 0$ Theorem 2.6 gives $2\lceil w/2 \rceil$, which is w for even w and $w + 1$ for odd w ; for $x_0 = 1$ Theorem 2.7 gives $q - 1$, which is $w + 1$ for even w and w for odd w . Hence the maximum horizon index is $w + 1$, so the maximum prefix length is $w + 2$. It is attained in the one case when w is even and in the zero case when w is odd, with the corresponding counts 2^w and $2^w - 1$. $\square$

The Rule-30-specific step is the OR latch in the proof of Theorem 2.2: the same finite row under Rule 90 keeps a zero center forever, so Corollary 2.5 is not a consequence of left permutivity. This is the Rule 90 filter passing on a proven result, and the template for reading every later obstruction.

3 Part II: The period-two problem — reduction, census, and a lead

3.1 The reduction chain

Period-two exclusion reduces, through a chain of proved uniform implications, to a single open separator. Writing (PT2) for “no nonzero finite configuration has $\text{Tr}_0(y) = \text{Tr}_0(F^2y)$,” the chain is

$$\gamma(n) \geq 1 \ \forall n \Rightarrow H_r(n) = \emptyset \Rightarrow \text{RW} \Leftrightarrow \text{DLP} \Rightarrow \text{SEP} \Rightarrow (\text{PT2}), \quad (6)$$

in which every arrow is proved and only the terminal separator SEP is open. Consequently a proof of $\gamma(n) \geq 1$ for all n would yield the first per-period exclusion theorem for the Rule 30 center column; the finite range is already discharged (below).

3.2 The forced-continuation census

A source word $W \in \{1, 2\}^n$ is extended by a deterministic forcing to a continuation that must remain hard-core; $\text{max-surv}(n)$ is the most forced rows any W survives and $\gamma(n) = (n + 2) - \text{max-surv}(n)$.

Proposition 3.1 (Census, exhaustive and verified). *$|H_r(n)| = 0$ exhaustively for $n = 1, \dots, 24$, all residues and both constants; the rotated-wedge SAT encoding is UNSAT to $n = 28$; and $\gamma(n)$, computed by a fast dedup method cross-checked against brute force through $n = 24$ with zero mismatches, satisfies $\gamma(n) \in [7, 14]$ for $17 \leq n \leq 30$ with no monotone trend.*

Remark 3.2. An earlier reading of the even- n subsequence as “climbing” was an artifact of sampling only even n ; the interleaved series has no trend. The single $n = 28, c = 2$ cell that suggested a fall was bracketed by its neighbors as a spike and is quarantined from all conclusions.

3.3 A structural lead: extinction is a needle, not a fog

Observation 3.3 (Terminal-class collapse; exhaustive $n \leq 18$). The words attaining $\text{max-surv}(n)$ number between two and eighteen, fall into one to four continuation classes, and within a class share a common suffix of length $n - O(1)$, produce the identical forced continuation, and fail on the identical symbol. The forcing’s live state is identical across a class (or differs only in the newest one or two cells). The phenomenon persists under restriction to hard-core sources, where the extremal words are exactly the clean periodic adversaries.

Two supporting facts are proved: a suffix lemma (words agreeing on a suffix from index m and on the first $2m$ forced symbols share their continuation forever; verified with zero exceptions) and that the sole Rule-30-specific ingredient of the forcing is the collapse $\Phi(2, \cdot) = \Phi(3, \cdot)$ of two left-actions, which is vacuous for Rule 90.

Conjecture 3.4. For all n , $\gamma(n) \geq 1$; equivalently, no reachable forcing state at source length n survives $n + 2$ forced rows. The extremal states, $O(1)$ per length, are the whole content.

3.4 Why first-moment and spectral bounds cannot close it

Proposition 3.5. *The hard-core transfer matrix has spectral radius exactly $\varphi = 1.618\dots$, so the induced bound $\lambda < 2$ holds vacuously at every width and cannot witness extinction (which needs $\lambda < 1$). The bound is tight for $k < k_\star$ and off by a factor ≈ 300 at extinction, so no finite-width transfer or first-moment argument closes the census; the departure past $k_\star$ is the whole phenomenon.*

4 Part III: The periodicity ladder and its limits

The ladder encodes “the center column is eventually p -periodic” as ω -automaton emptiness over column pairs, with the inverse transduction, a light-cone wedge, and a Buchi periodicity condition. Calibration reproduces the period-one theorems and passes the Rule 90 soundness control. For $p = 2$ the language is nonempty at every decided depth, and we prove this is not a computational limit but a structural one.

Theorem 4.1 (Three inert levers). *For the $p = 2$ ladder: (i) the accepting verdict is uniform in the right depth R over a range of raw sizes exceeding $5000\times$; (ii) the boundary pin is sandwiched, $\text{plain}(R+1) \subseteq \text{pin}(R) \subseteq \text{plain}(R)$, so it decides nothing depth does not; and (iii) the accepting tail system’s bisimulation-class count is exactly constant in the left depth k (15, 20, 32 in the three encodings, over a $3068\times$ range of raw states), because every accepting state lies in the saturated region where the left-depth constraints are inert. No lever of the ladder can force emptiness.*

Proposition 4.2 (The witness-extension test is invalid). *Extracting a ladder witness and testing whether it extends to a genuine lone-seed diagram cannot decide anything: on the Rule 90 control, where a genuine witness provably exists, the sampler misses it by a factor of four in depth. The nonemptiness is attributable to the free boundary of the truncated strip (obstruction F).*

5 Part IV: The register of obstructions

Ninety-four distinct routes have been attempted against Problems 1–3 in this project.¹ Nine obstruction classes account for essentially all of them. We state each as the reason a route dies, name the kill-reason type, and give its Rule 90 status. The exhaustive per-route register is the supplement.

- A. The $O(\log t)$ propagation wall.** Trace-anchored analysis determines a region of depth only $O(\log t)$, from one-bit-per-step information loss; three independent representations hit it. *Type:* depth. Does not bind the period-two counting question, which uses the periodicity hypothesis globally.
- B. The Rule 90 filter.** Any argument using only left permutivity, additivity, or expansivity proves a false statement about Rule 90. *Type:* rule-generic. The organizing obstruction of the whole paper.
- C. Single-column blindness.** Entropy- and complexity-boundary framings measure a two-dimensional field insensitive to any one column. *Type:* population/observable. Kills the Kolmogorov and block-entropy routes.
- D. The missing composition law.** Every proposed Problem 3 shortcut needs a composition law it never supplies. *Type:* composition.
- E. The measure-zero single-orbit gap.** Rule 30 is ergodic for Bernoulli measure, so almost-everywhere density results never see the lone seed. *Type:* measure. Kills the density routes to Problem 2.
- F. The free boundary of a fixed-depth strip.** Any fixed-depth column window leaves its outermost column unconstrained, and the phase slip relocates there. *Type:* boundary. The ladder’s wall (Part III).

¹Seventy-six in the primary register (PATH.md §7.1) plus eighteen merged from a second attack stream (experiments/overnight-arms/frontier_attack/FINDINGS.md §3, rows 73–90). One row number, 76, is shared by an unrelated route in each stream (S-adic subword complexity in the primary register; the d -uniform survivor invariant in the second); both are retained under their original numbers rather than renumbered, and are distinguished here by stream.

G. Arbitrary-input measures versus a fixed point. Circuit size, decision depth, sensitivity and the like are identically zero on one fixed input. *Type:* object-mismatch. Kills the Problem 3 complexity routes.

H. Finite certificates are complexity lower bounds, never more. Any bounded exclusion or census bounds a complexity function and cannot become an all-length theorem. *Type:* intrinsic. Why no census closes a theorem.

I. The pinned light-cone CNF. With n fixed, the circuit CNF refutes only that instance. *Type:* object-mismatch. Kills the pinned Problem 3 SAT.

Remark 5.1 (The five sightings of the needle). The collapse of Observation 3.3 was recorded five times before, under five names, and dropped each time for a kill reason of type C or E — a population statistic — when the deciding object was never the population but the $O(1)$ extremal states. This is the methodological lesson of the register.

Table 1 is the compact register: all ninety-four routes, one line each. Rows marked $\dagger$ come from the second attack stream (`experiments/overnight-arms/frontier_attack/FINDINGS.md` §3, numbered 73–90 continuing the primary register); row 76 is duplicated across streams by an unrelated numbering collision and both entries are kept under their original number. Full paragraphs, with exact quantities and file citations, are the accompanying supplement.

Table 1: The obstruction register: ninety-four routes.

#	Route	P	Status	Kill reason
1	R1: zero-set obligation (c eventually periodic $\Rightarrow r$ eventually periodic on $\{c_t = 0\}$)	1	OPEN	Best route; kill condition not fired.
2	R2: cascade descent from the pin	1	KILLED	Own kill condition fired: 1.14% of cells determined, density halves leftward, zero at $x \geq 1$.
3	R3: bounded-depth pin bootstrap (run-of-ones wedge)	1	KILLED	Proved dead before remeasurement: bounded one-runs give only a bounded wedge, not a descent.
4	R4: right-boundary-anchored arguments	1	KILLED	Right-boundary regularity reaches only $\sim 2.4 \log_2 t$ diagonals against a target at diagonal t .
5	R5: left permutivity / expansivity alone	1	KILLED	Rule 90 filter: left permutive, lone-seed centre column eventually periodic (Kopra).
6	R6: complexity/entropy boundary framings	1,2,3	KILLED	Literal version false in two lines; statistical rescue fails three ways; 2D entropy is column-blind.
7	R7: periodicity ladder, per-period exclusion	1	STALLED	$p = 2..8$ all NONEMPTY with verified witnesses; verdict uniform in depth, pin and Q .
8	R8: invariant-measure classification of orbit closure	2	OPEN	Consistent with unique ergodicity at every cell; not killed, not advanced.
9	R9: proof-complexity lower bound for c_n	3	OPEN	Precondition probe ran (GMUS near-constant fraction); inverted outcome did not fire; bound unattempted.
10	Arm 1: Lean/evolutionary search for a P1 proof	1	NOT RUN	Declined: “find a lemma” is not pre-registerable.
11	Arm 2: search for the density-1/2 invariant measure	2	NOT RUN	Declined: pathway lands one measure-zero step short of the lone seed.

#	Route	P	Status	Kill reason
12	Arm 3: WASM tournament for sub-quadratic centre algorithm	3	STALLED	Kill fires only past budget, search never reached it; recovers the known $O(n^2)$ exponent only.
13	Arm 3 iter. 2: GF(2) seed grammar + Z3 rewrite	3	NOT RUN	Design only; peephole rewrites cannot connect $\Theta(n^2)$ composition to sub-quadratic.
14	Arm 4: AIR / frequency-domain shortcut	3	KILLED	Pre-registered kill fired on first run: maximal linear complexity, exact ANF degree, plateauing density.
15	Arm 4 iter. 2: additive-FFT over $GF(2^k)$	3	NOT RUN	Design only; no root-of-unity structure in a binary field; no FiniteField sort in the SMT solver used.
16	Arm 5: Rule 150 kernel + sparse-error superposition	3	KILLED	$N_{\text{eff}} \sim n^{1.68}$, decisively superlinear; escape hatch (row/block parity) also measured shut.
17	Arm 6: finite 2-kernel / bounded-rank digit query	3	KILLED	Both pre-registered kills fired: ≥ 8191 residual sequences, GF(2) rank 512 at depth 9.
18	Arm 7: exact dyadic space-time tile grammar	3	STALLED	Diagnostic structure only; tile perimeters carry $\Theta(\text{scale})$ bits; five-point contract unmet.
19	Bounded Crosstalk search: dyadic tile composition algebra	3	KILLED	Zero surviving candidates across six critic rejections; synthesis over-reached.
20	Arm 8: ROBDD of the arbitrary-input transfer function	3	KILLED	Exponential growth ($\sim 1.88\times/\text{step}$) in all five tested variable orders.
21	Bounded Crosstalk search: single-seed transfer monoid algebra	3	KILLED	Truncated $GF(2)[x]/(x^k)$ candidate falsified: first mismatch at step 5.
22	Crosstalk elimination run 1	3	KILLED	Zero surviving candidates; four of five depend on a parity cancellation already measured false.
23	Crosstalk elimination run 2, typed exclusions	3	KILLED	Empty frontier; three rejections each needing an unproved or undefined lemma.
24	Crosstalk elimination run 3: carry-polynomial transfer	3	KILLED	Deterministic probe found scalar wrong at $n = 2$: candidate 1, ground truth 0.
25	Zero-trace fiber classification	1	PROVED	No nonzero finite configuration has an identically-zero centre trace; sharp horizon $2\lceil w/2 \rceil$; false for Rule 90.
26	All-one trace fiber	1	PROVED	All-one trace forces the checkerboard left half; with row 25, excludes every eventually-constant centre.
27	Phase-labelled generalization of the OR-latch	1	KILLED	Falsified at radius one: centre-one phase removes rather than preserves the right-neighbor latch.
28	Inverse-trace continuation	1	STALLED	Exact but has not produced a period-independent contradiction; longest agreement breaks at depth 166.
29	Periodic-mask contraction	1	KILLED	Falsified for every period word containing a zero: discrepancy moves earlier, never erased.
30	Bounded Crosstalk: period-uniform inverse-trace obstruction	1	KILLED	Zero of 72 claims verified; synthesis empty; two Lean artifacts contained sorry .
31	Same-forward-orbit collision reduction	1	OPEN	Equivalence proved; remaining obligation (no collision at any period) still needed, starting at $p = 2$.
32	Incremental SMT exhaustion of nonconstant periods	1	STALLED	Exact horizon table produced but no stable one-bit conflict pattern in the UNSAT cores.

#	Route	P	Status	Kill reason
33	Right-cone bit-parallel reconstruction + support certificate	1	STALLED	1824 sweeps, every one forces a one; no descent, obligation never shrinks to a finite check.
34	Right-cone diagonal periodicity	1	KILLED	Periodic region reachable from the boundary is $\sim 2.4 \log_2 t$ diagonals wide; centre is at diagonal t .
35	Extending the constant-trace mechanism to nonconstant periods	1	KILLED	Falsified by direct measurement: forced bit still differs at depth 512 for four period words.
36	Periodicity bridge: Kopra's width-two theorem to nonlinear class	1	KILLED	Of eight quiescent left-permutive ECAs, the nonlinear ones with full support reduce to Rule 30 itself.
37	Periodicity bridge: finite-delay reconstruction	1	KILLED	Explicit counterexamples at $p = 148, t = 1855$ and $p = 110, t = 13219$ with persistent adjacent-column defects.
38	Periodicity bridge: centre-trace injectivity	1	KILLED	$\{0\}$ and $\{0, 1\}$ share a centre trace at every checked step; defect escapes rightward.
39	Periodicity bridge: doubling separation	1	KILLED	First fails at $p = 4$, where both values are one.
40	Bounded Crosstalk: period-independent contradiction	1	KILLED	Zero candidates survived; ten rejects reduce to six recurring reasoning errors.
41	Alternating-trace fiber: bounded-depth exclusion certificates	1	OPEN	Closed-form survivor automaton found, but certificate levels grow with d and cannot reach uniformity alone.
42	Overnight literature-novelty triage (ten techniques)	infra	INFRA	Verdict table gating the overnight arms: 5 closed, 5 known-adjacent, 0 novel-open.
43	ANF/Walsh analysis of the iterated centre-bit function	—	PROVED	$\deg f_t = 2t - 1$ for all $t \geq 3$; implies nothing for P1/P2, provably inert for P3.
44	Classification of jointly invariant Markov measures, memory ≤ 3	2	STALLED	Decisive at $m = 1, 2$; Gröbner basis kill condition fired at $m = 3$ (110 min, no termination).
45	Anashin 2-adic ergodicity	2	KILLED	Four fatal reasons, including: Rule 90 has identical structure, so the object does not separate the rules.
46	Ore-relation ladder toward non-2-automaticity	1	OPEN	Reduced, not killed: each rung one-sidedly proved; stopped at one lemma needing a polynomial bound.
47	Single-seed density arm (martingale/counting/invariance)	2	KILLED	Original mechanisms structurally invalid (no probability space on a deterministic orbit); two weak byproducts survive.
48	Persistent homology / TDA on the spacetime point cloud	1,2	KILLED	Time-filtered H_1 bars biject with known white triangles; column periodicity constrains no neighborhood.
49	Quantum circuit mapping, entanglement entropy	1,3	KILLED	Single-seed basis-state entropy across the centre cut is identically zero.
50	Hydrodynamic / coarse-grained PDE limit	2	KILLED	Coarse-grained density is a spatial-average object; P2 is about one measure-zero orbit.
51	Non-standard analysis / hyperfinite rows	2	KILLED	Conservative: needs the standard equidistribution lemma already, and applies verbatim to Rule 90.
52	Homotopy type theory / loop-space framing	1	KILLED	All types are hSets; the statement 0-truncates to the plain first-order sentence.
53	Latent-space geometry to conserved window functions	1,2	NOT RUN	Never pre-registered; ranked third of the open triage rows.
54	Ladder tightening: right-wedge constraint	1	KILLED	Constraint already imposed in the code; every verdict NONEMPTY with or without it.

#	Route	P	Status	Kill reason
55	Ladder tightening: boundary pin	1	KILLED	Genuine constraint, prunes 39–58%, but rung 1 proved $\text{plain}(R+1) \subseteq \text{pin}(R) \subseteq \text{plain}(R)$: buys less than one column.
56	Ladder tightening: bounded-universal Q	1	KILLED	NONEMPTY to $Q = 14$; constraint and available freedom both grow as 2^Q .
57	Hyperbolic lightcone geodesic embedding	1	KILLED	Column-blind: overwriting column 0 moves the statistic by 10^{-4} – 10^{-7} against a 96% positive control.
58	Curvature of a multilinear relaxation	1,2	KILLED	Column-blind to machine epsilon; also a category error (no Riemannian metric on a max-plus semiring).
59	Spectral dimension of a Dirac operator on the shift algebra	1	KILLED	Column-blind, $O(1/W)$ to three digits; Rule 90 induces the same algebra with the opposite P1 answer.
60	OTOC / spectral form factor	1	KILLED	Premise false: Rule 30 is non-invertible, so no unitary evolution operator or RMT ensemble exists.
61	2-adic/Mahler expansion, unique ergodicity under Haar measure	2	KILLED	Hypothesis false in one line: δ_0 and uniform Bernoulli are both invariant.
62	Braid-group word length of defect world lines	1	KILLED	No braid word exists: Rule 30 defects merge and annihilate (forced by non-invertibility).
63	Algebraic immunity of f_t as an AC0 lower bound	3	KILLED	AC0 claim false: AND_n has maximal degree, algebraic immunity 1, and a depth-2 AC0 circuit.
64	AC0/Smolensky size bound from the ANF degree of f_t	3	KILLED	Refuted by the same AND_n counterexample; also the trivial regime ($n = 2t + 1$ variables).
65	Straight-line-program length along the single-seed line	3	KILLED	Category error: on one fixed point every measure of this kind is identically zero.
66	Sensitivity $s(f_t) \geq t$ along the single-seed path	3	KILLED	Numerically false: measured sensitivity sits below t and below the random-input maximum throughout.
67	Ground-truth centre-column generator and OEIS gate	infra	INFRA	Bit-parallel generator matched against OEIS A051023; gates every arm’s pipeline.
68	Modal compute charter and admission guard	infra	INFRA	\$200 ceiling, fail-closed guard; never entered (every result records \$0).
69	Crosstalk elimination run-book	infra	INFRA	Turns each failed mechanism into a typed exclusion later candidates must differ from.
70	Overnight autonomous run: prompt and fences	infra	INFRA	Fenced session producing rows 42–47; compliance verified by mtime, one disclosed exception.
71	Settled-knowledge digest for downstream sessions	infra	INFRA	Read-only survey of what is proved, dead, and off-limits.
72	Zero-tail note: manuscript, figures, Lean/SMT certifies	infra	INFRA	Packaging for row 25 plus a hostile proof audit; novelty over Jen 1986 confirmed componentwise.
73 [†]	Preimage-tree / backward-dynamics descent	1	KILLED	Column-blind at step 0; premise was a boundary confusion; object is the pin restated.
74 [†]	Ore ladder uniformity by induction on rungs	1	KILLED	Explicit family has strict Ore order at every level; only the induction sub-route dies (row 46 stays open).
75 [†]	Stroboscopic descent for $p = 2$	1	OPEN, no descent	Support grows exactly 2 per side under F^2 ; no width-shaped quantity can decrease.
76	S-adic decomposition + Morse-Hedlund complexity	1	KILLED	Killed by obstruction H; bound corrected upward $\sim 3,100\times$ on remerge (linear in prefix length, not $n_{\max}$).

#	Route	P	Status	Kill reason
76 [†]	Row 41 d -uniform survivor invariant (second stream)	1	OPEN, one bridge removed	Time-reindexed recurrence found, but natural truncation is vacuous by a light-cone argument.
77 [†]	Ladder realizability / half-plane slip	1	PROVED (limitation)	Slip extends to a full half-plane; ladder mode (i) can never return EMPTY at any depth.
78 [†]	P2 nontrivial density bounds	2	KILLED	Gap bound is linear in t ; already implied by row 47’s Theorem A.
79 [†]	Orbit-closure patch census	2	REFUTED (conditional)	Checkerboard is a Rule 30 temporal fixed point; R8’s target fails if it lies in Y (measured to $K = 6$).
80 [†]	Rowland-Yassawi nonlinear analog	2	NEGATIVE	Only arrow of the R-Y chain that avoids linearity delivers just a simplex.
81 [†]	Resolution lower bound for c_n	3	KILLED	Metric mis-specified (obstruction I).
82 [†]	Trace subshift soficness	1,2	NEGATIVE	Width-1 object is the full shift by construction (obstruction B); width-2 reduction gives only a lower bound.
83 [†]	Conserved non-additive window functions	1,2	KILLED	Column-blind width-independently; exhaustive absence to $w = 12$; Rule 90 gives identical answers.
84 [†]	Jen 1986 provenance	infra	INFRA	Clause (ii) obtained from citing sources; full 1986 text not retrieved; dependency audited to zero.
85 [†]	Sub-quadratic derivation family for c_n	3	KILLED	R9’s inverted kill did not fire; best family is exact $\Theta(n^2)$, $2.95\text{--}3.88\times$ under baseline (obstruction I).
86 [†]	Nonabelian conserved transfer invariant	1,2	KILLED, filter corrected	1,988 real invariants found, then closed generally: any monoid-valued transfer is $\leq \log_2 M $ bits about the orbit.
87 [†]	Polynomial Calculus degree for c_n	3	KILLED	Degree exactly 2 (Rule 30) vs. 1 (Rule 90), both constant in n ; fails the Rule 90 filter honestly (obstruction I).
88 [†]	2-adic Newton-polygon / valuation argument	1	KILLED	No closed functional equation exists (the pin’s nonlinear term is a Hadamard product, not a ring operation).
89 [†]	Cryptanalytic bias detection on the lone-seed trace	—	NO BIAS FOUND	BM complexity, correlation attack, NIST STS, autocorrelation to 2^{21} ; smallest corrected $p = 0.627$.
90 [†]	ML-guided conjecture mining	1	REFUTED at depth 20000	All 30 surviving candidates break within 38 steps of the training boundary; only the known OR-latch pin survives.
91	OpenEvolve LLM-driven search for sub-quadratic $c(n)$	3	NOT RUN (blocked)	Sanity gate failed on re-verification: wall-clock instrument credited the baseline with beating itself.
92	Deterministic fuel-counter evaluator (fixes row 91’s instrument)	3	INSTRUMENT VALIDATED	Word-RAM AST counter recovers exponent 1.99946 at $r^2 = 0.9999999$; publication claim withdrawn (prior art).
93	Bounded exact circuit synthesis for $n \rightarrow c(n)$	3	PROBED, OPEN	$k_{\min}(4) = 5$ unconditional; $k_{\min}(5) \geq 7$ conditional; strong outcome did not fire; P3 untouched.

6 Part V: Scope, the unifying fact, and open problems

Problems 1, 2 and 3 are open and nothing here changes that. Part I settles eventual period one completely, for a class strictly larger than the lone seed, and is the paper’s one unconditional advance

on the center-column question. Part III’s limitation theorems close a specific mechanized route to period two. Part II’s census and lead delimit period two without deciding it. The unifying fact is the Rule 90 filter: the OR nonlinearity is the entire difference, every proven result routes through it, and every failed route does not.

The concrete open leads are three: whether the extremal forcing states of Observation 3.3 form a family across n (the untried comparison, which if affirmative brings a fixed-family cocycle argument into range); a Rule 90 carry-kernel census as the missing soundness control on the period-two thread; and the separator SEP itself, the one open arrow of (6).

7 Reproducibility

All computational claims are reproduced by scripts in the accompanying archive, seeded and cross-checked; the exhaustive audits and SMT certificates supplement but do not replace the unbounded proofs of Part I. The full corpus — scripts, logs, and the result documents underlying the register — is deposited separately (Zenodo DOI: [to mint]), and the exhaustive per-route register is the supplement.

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